

Machine Learning

Random Forests



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Machine Learning (25SC2107E)

Session Outcome

- Understand why one decision tree is unstable, and how many trees — *bagging, feature subsampling, out-of-bag error* — become a random forest.

Key Terms — Bias, Variance and Ensembles

Definitions

Bias — the error that comes from a model being too simple to represent the truth. A high-bias model is wrong in the same way no matter which data it is trained on.

Variance — the error that comes from a model being too sensitive to the particular rows it was trained on. A high-variance model changes a lot when the data changes a little.

Overfitting — fitting the noise in the training set, so training error is very low while test error stays high.

Pruning — cutting branches off a grown tree (or stopping it early) to reduce variance, at the cost of some training accuracy.

Ensemble learning — combining the predictions of several models into a single answer.

Base learner — one individual model inside an ensemble (here, one decision tree).

One Tree Learns the Training Set by Heart

Full tree, 85 leaves

train



1.000

test



1 in 4 wrong

0.738

Pruned tree, depth 5

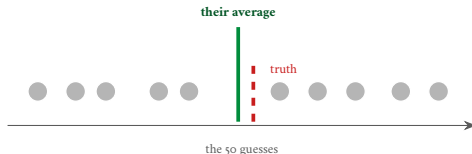
test



0.767

- Pruning helps a little — but we pay for it with training accuracy.

Can we keep the deep tree and improve testing accuracy?



Every guess can be wrong, but their average is approximately right.

$$\text{Var}(\bar{y}) = \underbrace{\rho \sigma^2}_{\text{how \u00a1ike the trees are}} + \underbrace{\frac{1-\rho}{B} \sigma^2}_{\text{how \u00a1any trees}}$$

- more trees $\implies$ higher $B \implies$ reduces variance
- different trees $\implies$ small ρ (correlation coefficient) $\implies$ reduces variance

The Random forest is about: *more* trees, and *more different* trees.

Bagging

Key Terms — Bootstrap and Bagging

Definitions

Bootstrap sample: a sample of n rows drawn from the n training rows with replacement, so some rows appear more than once and others not at all.

Bagging (Bootstrap aggregating): training one model on each bootstrap sample and combining them: majority vote for classification, average for regression.

Aggregation: the rule that turns many base-learner outputs into one prediction.

In-bag / out-of-bag (OOB) — for a given tree, the rows that were in its bootstrap sample, and the $\approx 37\%$ that were not.

Out-of-bag error — the error obtained by predicting each row using only the trees that never saw it. It estimates test error without holding any rows back.

BOOTSTRAP SAMPLING

draw n rows with replacement

ORIGINAL DATA

Row 1
Row 2
Row 3
Row 4
Row 5
Row 6

n rows,
each equally likely
to be drawn

SAMPLE 1

Row 1
Row 3
Row 3
Row 4
Row 6
Row 6

Row 3, Row 6 twice
Rows 2, 5 never drawn

SAMPLE 2

Row 2
Row 2
Row 3
Row 5
Row 5
Row 6

Row 2, Row 5 twice
Rows 1, 4 never drawn

SAMPLE 3

Row 1
Row 1
Row 4
Row 4
Row 5
Row 6

Row 1, Row 4 twice
Rows 2, 3 never drawn

 = a row drawn more than once

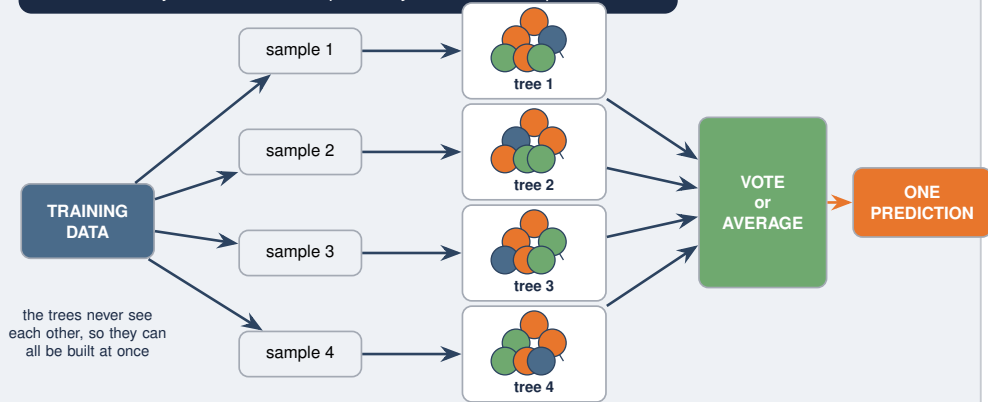
Every sample is the same size as the original, but not the same rows.

A given row is missed with probability $(1 - \frac{1}{n})^n \rightarrow e^{-1} = 0.368$

so each sample holds about 63% of the distinct rows — the other 37% are that tree's *out-of-bag* rows.

BAGGING = BOOTSTRAP AGGREGATING

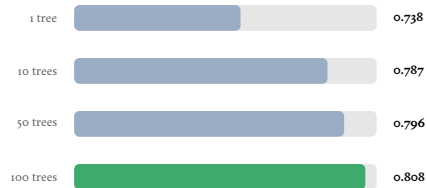
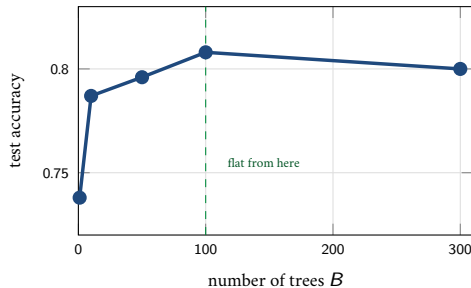
every tree is trained *independently*, on its own sample



Each tree is grown fully, with no pruning: deep tree = **low bias, high variance**.

Averaging many such trees cancels the variance — *classification* takes a majority vote, *regression* takes the mean.

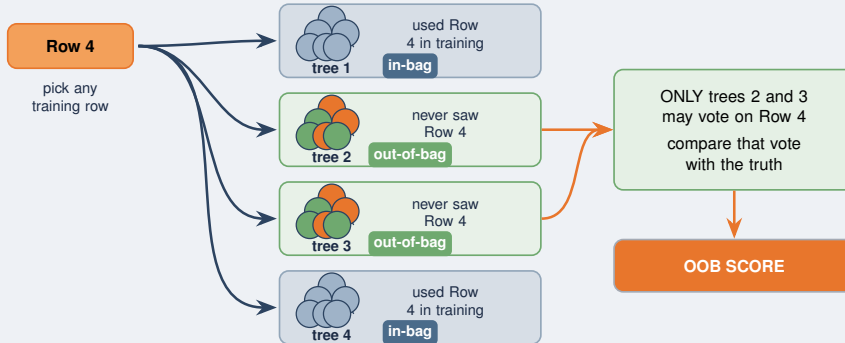
How Many Trees?



- One tree 0.738 $\rightarrow$ a hundred trees 0.808.

OUT-OF-BAG (OOB) ERROR

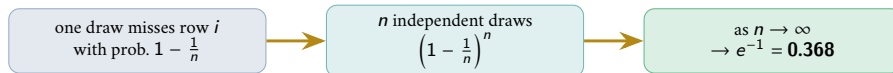
every row is a free test row for the trees that never saw it



Repeat for *every* row. A row is out-of-bag for about **37%** of the trees, so with $B = 500$ it gets a private jury of roughly 184 trees.

```
RandomForestClassifier(oob_score=True) → clf.oob_score_
```

Where the 37% Comes From



	fraction	for $n = 10,000$
rows used by a tree (in-bag)	0.632	6,321 distinct rows
rows never seen by a tree (OOB)	0.368	3,679 rows
trees for which one given row is OOB	$0.368B$	184 trees if $B = 500$

- Each row therefore has its own private jury of about $0.368B$ trees that never studied it.

Quick Check — Bootstrap and OOB

Questions

- 1 A bootstrap sample is drawn from $n = 1,000$ rows. About how many distinct rows appear in it?

Quick Check — Bootstrap and OOB

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Ans. $0.632 \times 1000 \approx 632$ distinct rows; the other ≈ 368 are out-of-bag.

- ❷ A forest has $B = 500$ trees. For one given training row, about how many trees are entitled to use it for OOB prediction?

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- ❸ Does raising B from 100 to 1000 change the 37%?

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- ❸ Does raising B from 100 to 1000 change the 37%?

Ans. No. The fraction is a property of one bootstrap draw. Raising B only enlarges each row's jury, which makes the OOB estimate less noisy.

Random Forests

Key Terms — Random Forests

Definitions

Tree correlation ρ — how alike the predictions of two trees in the ensemble are. $\rho = 1$ means identical trees, $\rho = 0$ means unrelated ones.

Feature subsampling — offering only m randomly chosen features out of p as split candidates at each node, with a fresh draw at every node.

m (max_features) — how many features are offered at a node. Defaults: $m = \sqrt{p}$ for classification, $m = p/3$ for regression.

Decorrelation — deliberately making the base learners less alike, in order to lower ρ and so lower the variance of their average.

Random forest — bagging of unpruned decision trees, plus feature subsampling at every node.

Bagged Trees Still Look Alike

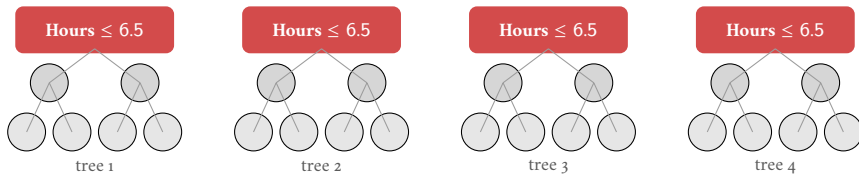


Figure: 7 of our 8 bagged trees asked the same first question

- One strong column dominates every bootstrap sample, so every tree copies it.
- Measured correlation between bagged trees: $\rho = 0.45$ — almost half the variance stays.

Fix : if the trees copy each other, *hide some columns* from them.

FEATURE SUBSAMPLING

at every node, offer only m of the p columns

NODE A



offered: **Hours, PrevScore, PhoneHrs**

NODE B



offered: **Attend, Sleep, Backlogs** — a *fresh* draw

Each tree gets slightly worse on its own,
because its best column is often hidden

But the trees become much less alike
correlation ρ falls $0.45 \rightarrow 0.33$,
and the *ensemble* improves

Defaults: $m = \sqrt{p}$ (classification), $m = p/3$ (regression).
Setting $m = p$ turns the forest back into plain bagging.

Quick Check — Feature Subsampling

Questions

- 1 A classification forest has $p = 36$ features. What is the default m ?

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Ans. $m = \sqrt{36} = 6$ columns, drawn afresh at every node — not once per tree.

- 2 Setting `max_features=None` ($m = p$) turns the forest into what, and what happens to ρ ?

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- ❷ Setting `max_features=None` ($m = p$) turns the forest into what, and what happens to ρ ?

Ans. Plain bagging. Every tree may see the dominant column at every node, so they all split on it, ρ rises, and the floor $\rho\sigma^2$ stays high.

- ❸ Smaller m makes each individual tree worse. Why does the ensemble still improve?

Quick Check — Feature Subsampling

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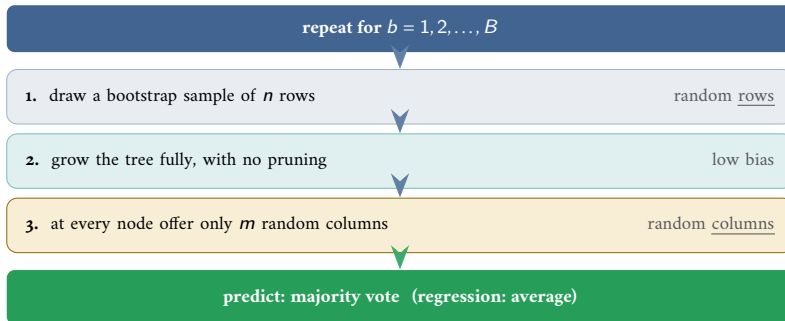
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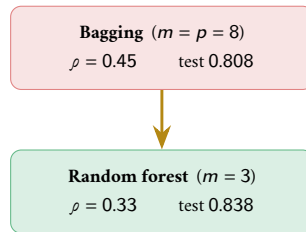
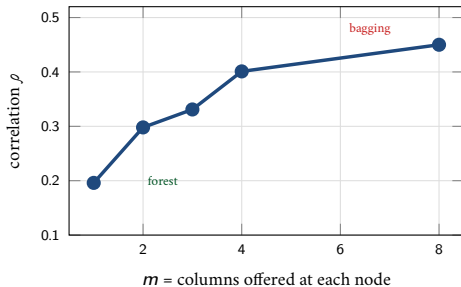
Ans. $\text{Var}(\bar{y}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$. Smaller m raises σ^2 a little but lowers ρ a lot, and ρ multiplies the term that never vanishes.

The Random Forest Steps



- Two sources of randomness: the rows (step 1) and the columns (step 3, fresh at every node).
- Only three settings matter: `n_estimators`, `max_features`, `min_samples_leaf`.

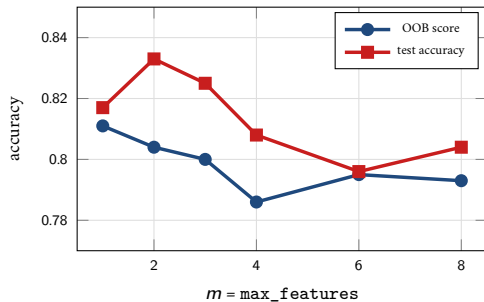
Fewer Columns $\Rightarrow$ Less Correlation $\Rightarrow$ Better



only m changed

Same data, same trees — hide a few columns and the score rises.

Choosing m



m small — very different trees,
but each one weak

$m = \sqrt{p}$ — the balance point

m large — strong trees,
but near-identical

- Here $\sqrt{8} \approx 3$.
- Choose m based on the *OOB score*, not based on the test set accuracy.

Scenario — Crop Disease Forecasting

Case Study

An agri-tech company forecasts crop-disease outbreaks for $N = 20,000$ field plots using $p = 36$ satellite and soil-sensor features. Single trees proved unstable across seasons, so the team deploys a random forest with $B = 500$ trees. No separate validation set was set aside.

- (a) Describe how offering only $m = \sqrt{p}$ features at each split reduces the correlation between trees compared with standard bagging, and explain how that leads to overall variance reduction.
- (b) Show that the probability of a specific plot being excluded from a bootstrap sample approaches $1/e \approx 0.368$ as $N \rightarrow \infty$. Using this, compute the expected number of out-of-bag plots per tree, and the number of trees that can score any one plot.

Case Study — Answer (a) and (b)

(a) Feature subsampling and variance

With all $p = 36$ columns offered, the single strongest predictor wins the root split in *almost every* bootstrap sample, so all 500 trees begin identically and stay alike: ρ is large. Offering only $m = 6$ random columns means the dominant feature is absent from roughly $\frac{5}{6}$ of the nodes, forcing other features to be used, so the trees are genuinely different and the ρ falls.

$$\text{Var}(\bar{y}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$$

The second term is already killed by $B = 500$. Only a smaller ρ can lower the *first* term, which is the floor as $B \rightarrow \infty$. Each tree does get slightly weaker (a modest rise in σ^2), but the fall in ρ dominates — so the ensemble improves.

(b) The out-of-bag count

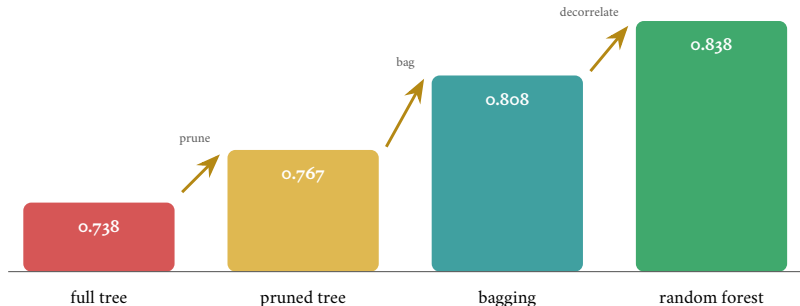
One draw misses a given plot with probability $1 - \frac{1}{N}$; the N draws are independent, so

$$P(\text{plot excluded}) = \left(1 - \frac{1}{N}\right)^N \xrightarrow{N \rightarrow \infty} e^{-1} = 0.3679.$$

OOB plots per tree: $0.3679 \times 20,000 \approx 7,358$ plots.

Trees that can score one plot: $0.3679 \times 500 \approx 184$ trees.

The Scoreboard: Testing Accuracy



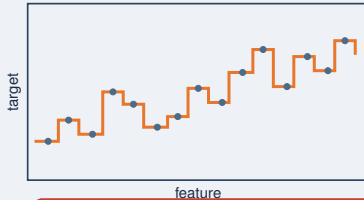
- All four reach 1.000 or near it on training data, but the forest generalises best.

Training accuracy tells you nothing. Only the test or OOB score does.

ONE DEEP TREE vs A RANDOM FOREST

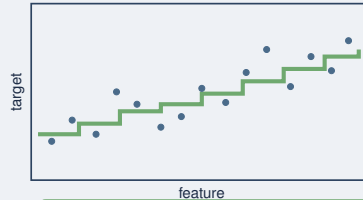
the same data, two very different fits

SINGLE DEEP TREE



High variance — a leaf
for almost every point.
Train 1.000, test 0.738

RANDOM FOREST



Lower variance — the noise cancels out.
Train 1.000, test 0.838

Both score 1.000 on the *training* rows, yet only one works on new rows. Training accuracy is not reliable!

Random Forest in Code

```
1 from sklearn.ensemble import \
2     RandomForestClassifier
3
4 rf = RandomForestClassifier(
5     n_estimators=300,
6     max_features="sqrt",    # m
7     min_samples_leaf=1,
8     oob_score=True,
9     n_jobs=-1,
10    random_state=1)
11
12 rf.fit(Xtr, ytr)
13
14 print(rf.oob_score_)        # 0.804
15 print(rf.score(Xte, yte))   # 0.833
```

- `n_estimators` — raise until the OOB score flattens.
- `max_features` — "sqrt" for classification.
- `min_samples_leaf` — raise to 5 if the data is noisy.
- `n_jobs=-1` — use every core.

Do not set `max_depth`. The trees are meant to grow fully.

Ensemble Learning

Key Terms — Ensemble Methods and Voting

Definitions

Hard voting — each model votes for a class *label*, and the label with the most votes wins.

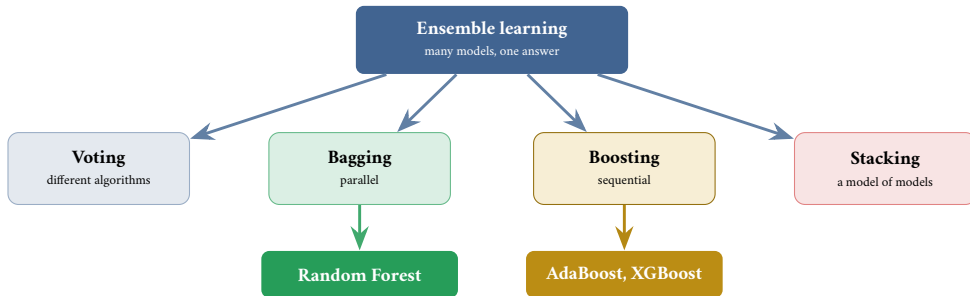
Soft voting — each model outputs a class *probability*; the probabilities are averaged and the highest average wins.

Bagging — independent models trained *in parallel* on resampled data and combined by vote or average. Reduces *variance*.

Boosting — models trained *in sequence*, each one correcting the errors of those before it. Reduces *bias*.

Stacking — training a small *meta-model* to combine the predictions of several different base models.

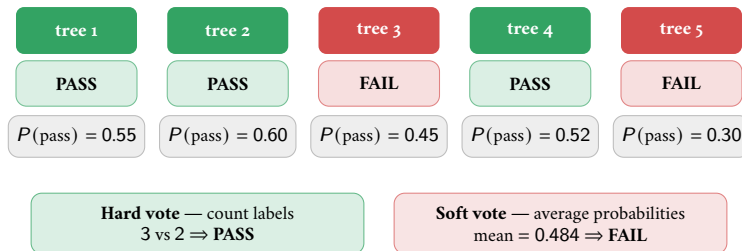
Where Forests Sit in the Family



next lecture

- **Bagging:** many strong models built independently, then averaged → decreases **variance**.
- **Boosting:** many weak models built in order, each fixing the last → decreases **bias**.

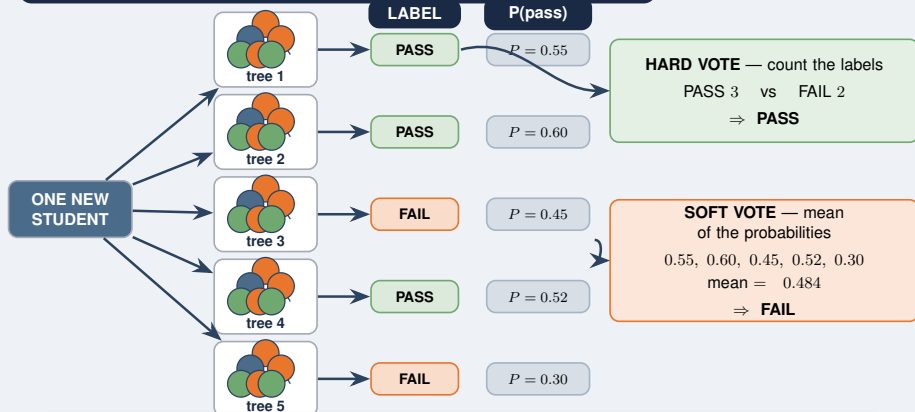
Hard Voting and Soft Voting



- Soft voting noticed the three PASS votes were barely above 0.5, while one FAIL vote was confident.
- Prefer **soft voting** when the models give sensible probabilities. A forest votes softly by default.

AGGREGATING THE PREDICTIONS

hard voting counts labels, soft voting averages probabilities



The two rules disagree here: soft voting noticed that the three PASS votes were barely above 0.5, while one FAIL vote was very confident.

Prefer **soft voting** when the models give sensible probabilities — a random forest uses it by default.

When to Use a Random Forest

✓ Good choice when

a strong baseline with almost no tuning

mixed, unscaled feature types

non-linear effects and interactions

a few thousand to a few million rows

✗ Use something else when

you must read the model → one pruned tree

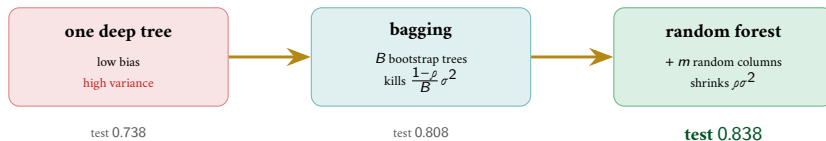
text, images, audio → neural networks

the truth really is linear → ridge / lasso

you must predict outside the training range

Three traps: pruning the trees, tuning m on the test set, and reading importances as causes.

Summary — Random Forests



- Bootstrap changes the rows; feature subsampling changes the columns.
- OOB error: the 37% of trees that never saw a row are its private test set — validation for free.
- Defaults that work: $B = 300$, $m = \sqrt{p}$ (or $p/3$ for regression), trees grown fully.

MCQs

❶ A bootstrap sample of size n drawn from n rows contains, on average, about
(a) 37% (b) 50% (c) 63% (d) 100% of the distinct rows

Answer: (c)

❷ Bagging mainly reduces
(a) bias (b) variance (c) noise (d) training time

Answer: (b)

❸ Trees inside a random forest should be
(a) grown fully (b) pruned hard (c) depth 1 (d) linear

Answer: (a)

❹ Out-of-bag error uses, for each row,
(a) the test set (b) all the trees (c) only the trees that did not train on it (d) a fresh bootstrap

Answer: (c)

MCQs

- 5 Setting `max_features = p` turns a random forest into
(a) a single tree (b) plain bagging (c) boosting (d) logistic regression
- 6 In $\text{Var}(\bar{y}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$, adding more trees reduces
(a) the first term (b) the second term only (c) both (d) neither
- 7 The default `m` for a classification forest with `p = 16` features is
(a) 16 (b) 8 (c) 4 (d) 1
- 8 Increasing `n_estimators` from 100 to 1000 will
(a) overfit (b) not overfit, only cost time (c) reduce bias a lot (d) raise ρ

Answer: (b)

Answer: (b)

Answer: (c)

Answer: (b)